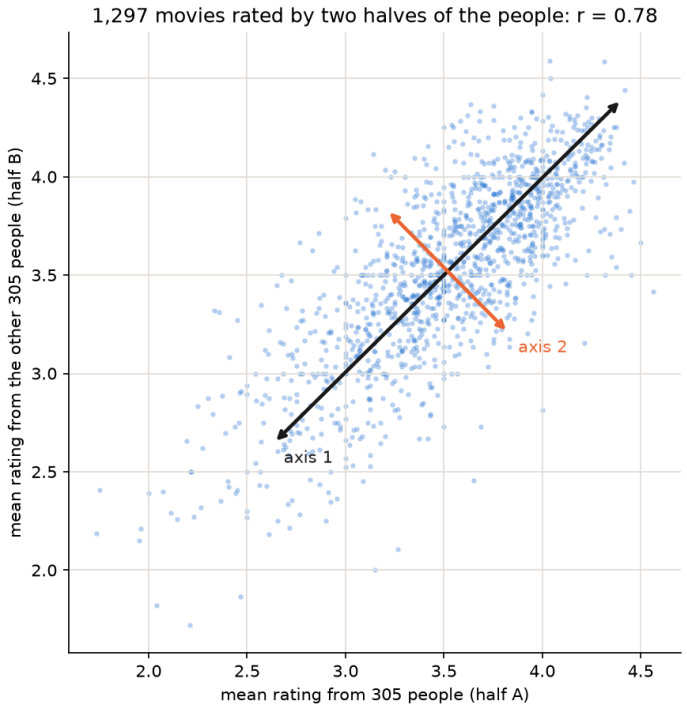
CS 301 · MEETING 14 · ACT 3

The Axes of a Cloud

Principal axes describe the spread of centred data. Covariance eigenvalues give the variance along each axis and determine rank and condition number.

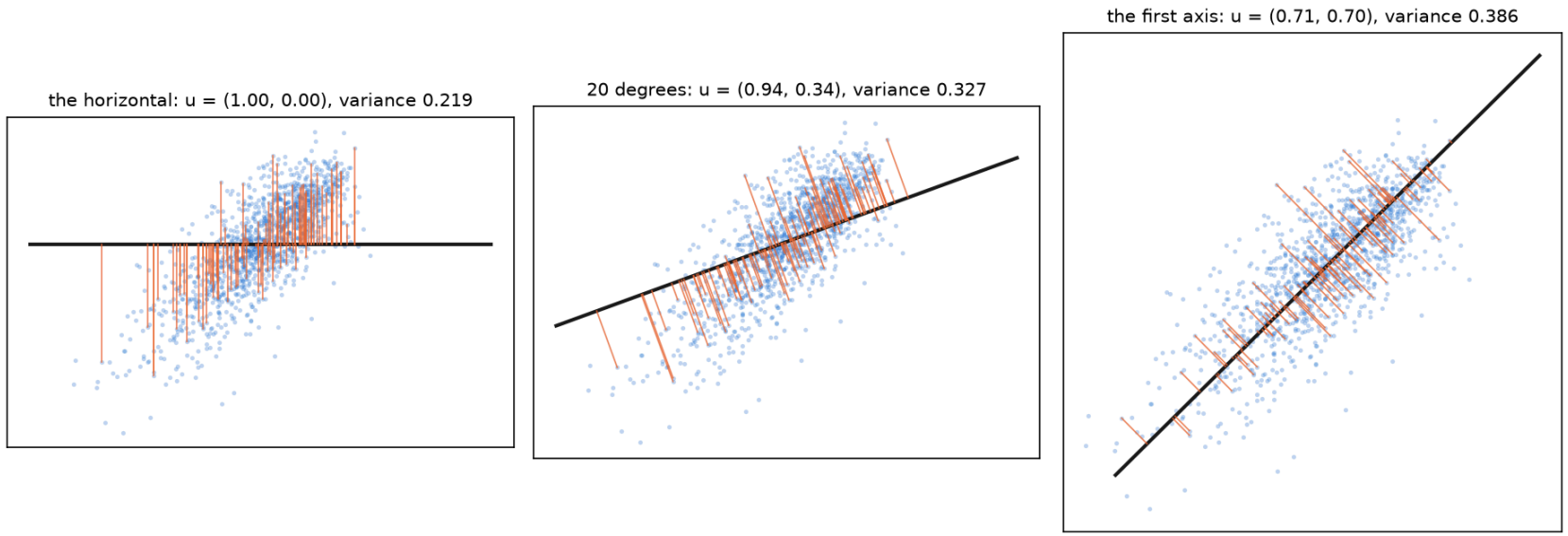
Mean ratings from two groups of users



- The 610 people split at random in two halves; a movie's x and y: its mean rating from each half
- 1,297 points; r = 0.78: the cloud leans on the diagonal

Projections onto a direction

$$p_i = (x_i - \mu) \cdot u, \quad \|u\| = 1 \qquad p = X_c\, u \qquad \mathrm{Var}(p) = \frac{1}{n}\sum_i p_i^2$$



- The dot product gives a signed coordinate along the line
- `np.var(Xc @ u)`: all 1,297 projections in one product, then their variance

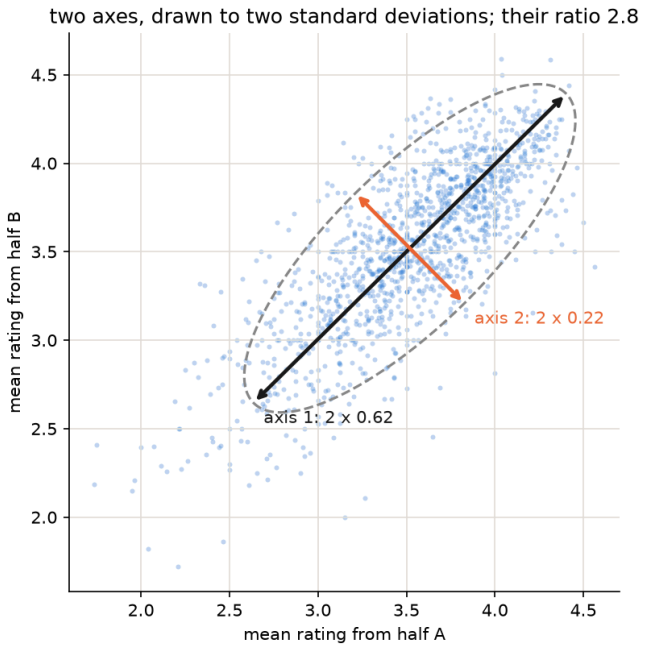
The first principal axis

```
axis 1: direction (-0.71, -0.70), singular value 22.4, variance of the projections 0.386
axis 2: direction (0.70, -0.71), singular value 7.9, variance of the projections 0.048
the ratio of the two axes: 2.84
for comparison, the variance of the projections on (1, 0), the horizontal axis: 0.219
```

```
Correct: the variance along the diagonal is 0.386, against 0.386 along axis 1, which the svd found at
(-0.711, -0.703): the diagonal is almost, not exactly, the best direction.
```

- The **first principal axis**: the direction u with the largest variance of the projections
- Here u ≈ ±(0.71, 0.70); either sign describes the same axis

Two axes, an ellipse



- The second axis: perpendicular to the first; the smallest variance
- Drawn to two standard deviations: 1.24 and 0.44 stars
- The **singular value** of an axis: the length of the vector of all the projections, $\sigma = \text{std} \cdot \sqrt{n}$

Axes from the covariance matrix

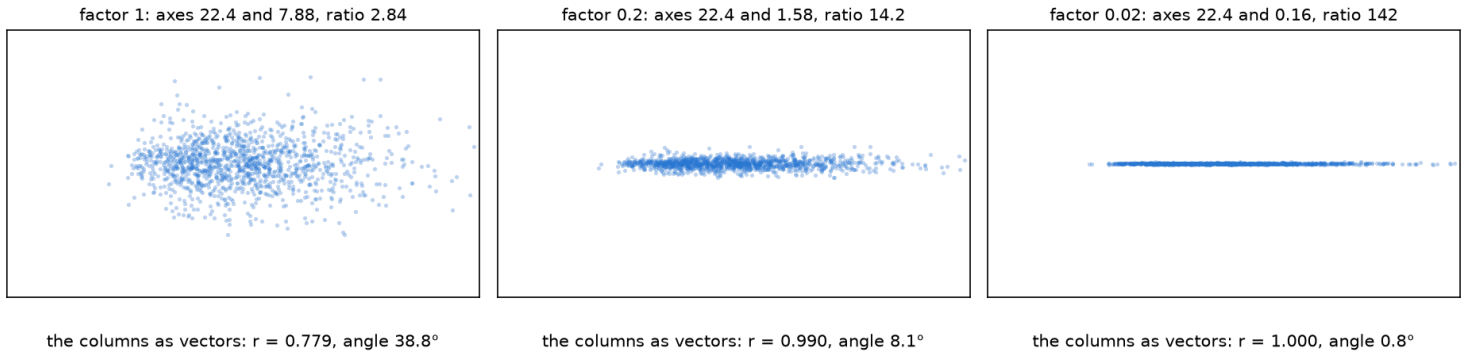
$$\text{Cov} = \frac{1}{n} X_c^T X_c, \qquad \text{Cov}\, v = \lambda\, v, \qquad \lambda_k = \frac{\sigma_k^2}{n}$$

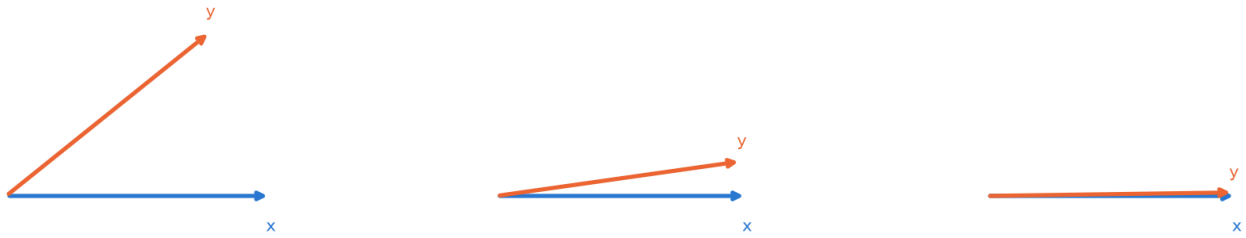
$$\kappa(X_c) = \frac{\sigma_1}{\sigma_2}, \qquad \kappa(\text{Cov}) = \frac{\lambda_1}{\lambda_2} = \kappa(X_c)^2$$

```
the covariance matrix Xc.T @ Xc / n:
[[0.219 0.169]
 [0.169 0.215]]
its eigenvalues: 0.386 and 0.048; its eigenvectors: (-0.71, -0.70) and (+0.70, -0.71)
the singular values of Xc, from np.linalg.svd: 22.37 and 7.88; squared and divided by n: 0.386 and
0.048
condition number of Xc: 2.84 = np.linalg.cond(Xc) = 2.84; of the covariance matrix: 8.05, its square
```

- The eigenvectors of the covariance matrix are the axes; its eigenvalues the variances along them
- `np.linalg.eigh(Xc.T @ Xc / n)`; `np.linalg.svd(Xc)` gives the same axes and the σ

Compressing the second coordinate





- Multiply the coordinate along axis 2 by a factor; nothing is re-centred or re-standardized
- κ(Xc), the ratio of singular values, grows; at factor 0 the rank drops from 2 to 1
- The columns x and y as two vectors: the angle whose cosine is r closes

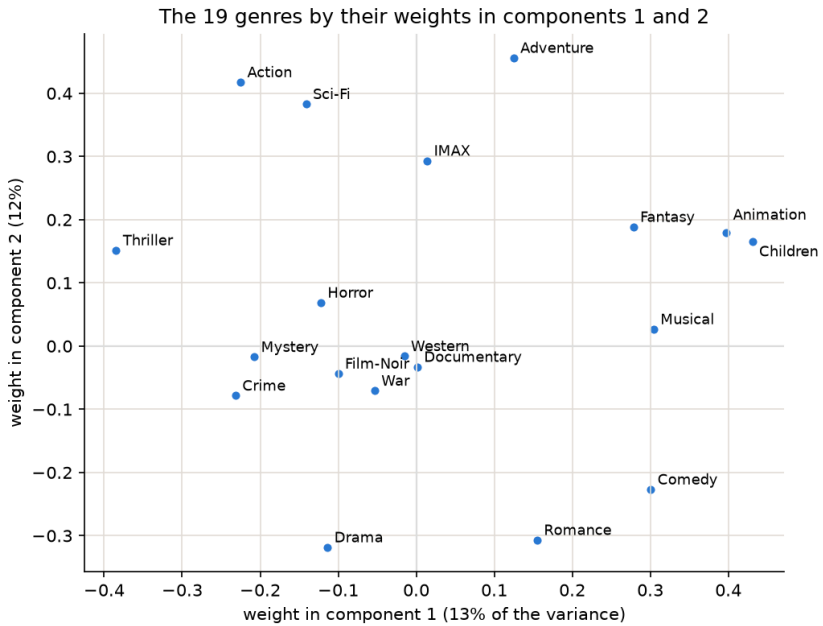
Principal axes of the genre data

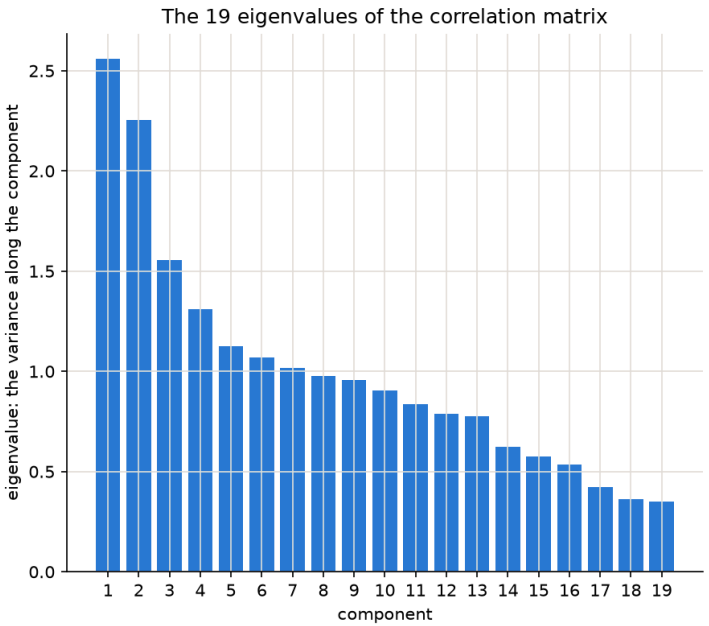
$$C = \frac{1}{n} Z^T Z, \qquad C\, v_k = \lambda_k\, v_k, \qquad k = 1, \ \ldots, 19$$

```
eigenvalues, eigenvectors = np.linalg.eigh(C)      # symmetric; increasing order
eigenvalues, eigenvectors = eigenvalues[::-1], eigenvectors[:, ::-1]  # largest first
share = eigenvalues / eigenvalues.sum()            # the variance share of each axis
```

- Each movie: 19 standardized genre coordinates; the cloud has 19 axes
- The axes: the eigenvectors of the 19 × 19 correlation matrix, the **principal components**
- The eigenvalues: the variances along them. Finding them is **PCA**

The genres on the first two components





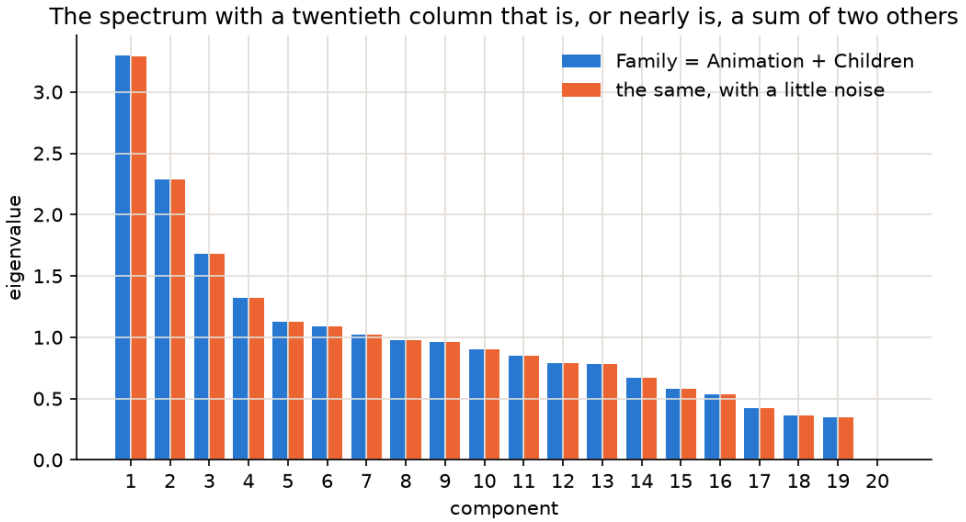

```
component 1: negative weights ['Thriller', 'Crime', 'Action', 'Mystery'], positive weights
['Children', 'Animation', 'Musical', 'Comedy']
component 2: negative weights ['Drama', 'Romance', 'Comedy', 'Crime'], positive weights ['Adventure',
'Action', 'Sci-Fi', 'IMAX']
components needed for half of the variance: 6 of 19; for 90%: 15
```

Rank and condition number

```
the largest eigenvalue 2.558, the smallest 0.349, their ratio 7.33
the square root of the ratio 2.707; np.linalg.cond(Z) = 2.707
the rank of Z: 19 of 19 columns
```

- The **rank** of Z: the number of nonzero eigenvalues: 19, every direction has spread
- $\kappa(C) = \lambda_{max} / \lambda_{min} = 7.3$; $\kappa(Z) = \sqrt{7.3} = 2.7 =$ `np.linalg.cond(Z)`
- A zero eigenvalue: a direction with no spread; a column that is a combination of the others

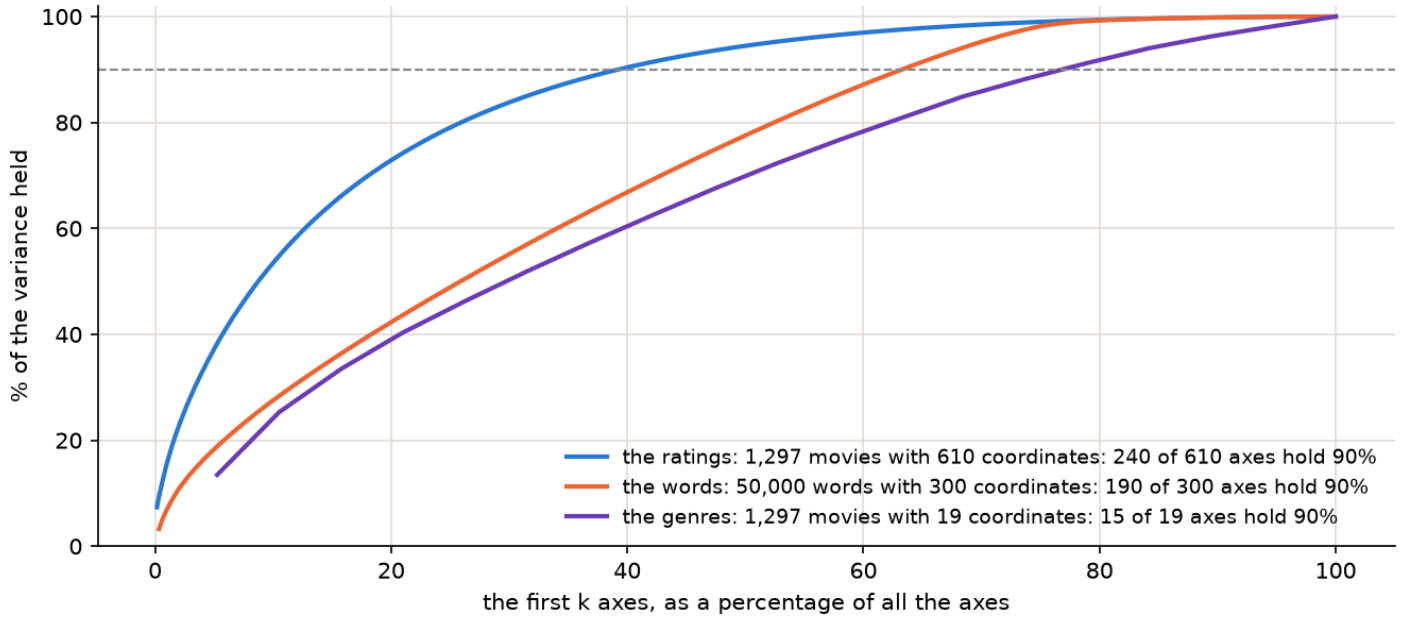
A dependent column: Family = Animation + Children




```
the smallest of the 20 eigenvalues: 1.7e-15, zero up to round-off; the numerical rank of Z20: 19 of
20 columns
with noise of standard deviation 0.05 on Family: the smallest eigenvalue 0.0058, the ratio 569,
np.linalg.cond(Z20n) = 23.9
```

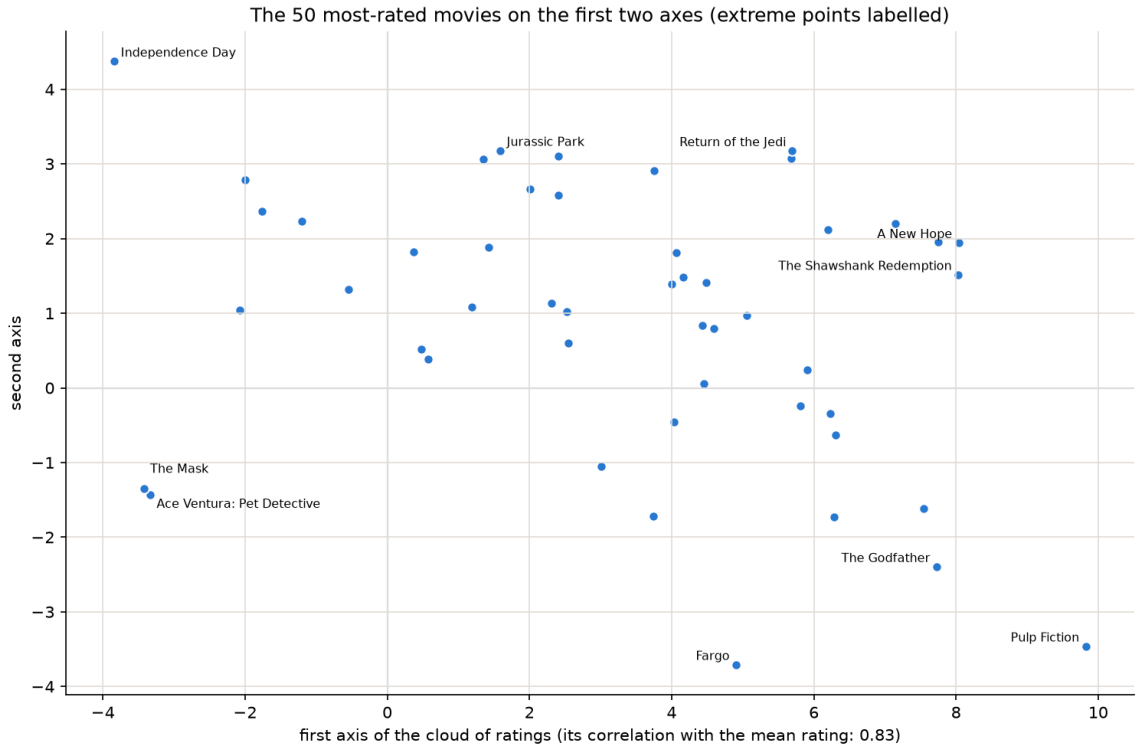
- A column that is the sum of two others: one eigenvalue 0 up to round-off; rank 19 of 20; exact κ infinite
- With noise: nearly a sum; the eigenvalue 0.006, $\kappa(C) \approx 570$, $\kappa(Z) = 24$

Variance captured by the principal axes



- Axes needed for 90% of the variance: 240 of 610 for the ratings, 190 of 300 for the words, 15 of 19 for the genres
- All three need many axes to retain 90% of the variance
- The Family example illustrates exact rank deficiency

The map of 50 movies



- Each movie: 610 coordinates, its column of `Rc`; the two longest axes of 1,297 points
- Axis 1 moves with the mean rating: r = 0.83 over all the movies
- The ratings determine the axes; signs set the display orientation

Matrix calculations for exploring data

the question	the matrix	calculation	read as
How do the features relate?	`Z` : columns standardized	`Z.T @ Z / n`	correlations
How similar are two columns?	`U` : unit-length columns	`U.T @ U`	column cosines
How similar are two rows?	`U` : unit-length rows	`U @ U.T`	row cosines
Where is the cloud spread?	`Xc` : columns centred	`eigh(Xc.T @ Xc / n)`	axes, variances, rank, κ

- Normalize the vectors being compared. PCA also requires an eigendecomposition.